



Condition controllable generation of 3D molecules using textual prompts by multimodal equivariant diffusion model



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Generating novel and functional molecules is an essential task in drug discovery, particularly in addressing the critical challenges of antibiotic resistance and the scarcity of effective treatments for major diseases such as cancer. Three-dimensional (3D) structure design can directly reflect a molecule's biological function, while its complexity and topology irregularity make de novo 3D molecule generation highly difficult under valid geometric constraints. Conditional or controllable design is a promising solution for this challenging task in a more accurate and quick expectation manner. In this study, we propose TDMol—a Text-guided De novo 3D molecule generation approach based on a multimodal diffusion model. TDMol designs a new two-stage modality alignment contrastive deep learning pipeline to extract the cross-modality shared knowledge and unique features from different modalities, including molecular textual descriptions and molecular 2D/3D structures, enabling conditional and controllable molecule generation. To the best of our knowledge, TDMol is the first to realize text-3D modality alignment for conditional molecule generation. Experimental results demonstrate that TDMol achieves a significant performance enhancement for generating valid and reliable 3D molecular structures. This work highlights the potential of multimodal foundation models in digital medicine, offering a scalable and generalizable framework for 3D molecule generation that can be adapted across diverse clinical settings. A user-friendly webserver of TDMol has been deployed for academic use at <http://www.csbio.sjtu.edu.cn/bioinf/TDMol>.

The landscape of clinical medicine is increasingly moving towards precision and personalization, demanding a new generation of therapeutics¹. Artificial intelligence-based drug design stands at the forefront of this transition and has recently attracted widespread attention worldwide. By enabling the de novo design of drug candidates for targets associated with major diseases such as cancer, where therapeutic options are scarce, it effectively enhances the precision and accelerates the progression of personalized medicine². Consequently, developing more powerful drug design technologies is paramount for bridging the gap between the basic research in medicinal chemistry and the creation of effective, targeted clinical interventions³.

Compared with protein drugs and nucleic acid drugs, chemical molecules have been well investigated because of their high binding affinity with the biological targets and high vivo stability⁴. Hence, generating novel, diverse, and synthetically accessible molecules with low toxicity and high drug-likeness is an essential task in drug design⁵. Most studies use the 1-dimensional (1D) and 2D features of molecules to model to generate the

2D structure of molecules⁶. However, the 3D structure is the most natural representation of molecules in biological processes such as drug-target interaction. Although the 3D structure of molecules can be inferred from its 2D structure by the density functional theory (DFT)⁷ or the prediction using machine learning methods⁸. The inference approaches bring additional errors and also lose the corresponding function of raw molecules, that is, the binding affinity with the biological targets⁹.

Recent deep generative models, especially diffusion models, have made considerable progress in 3D molecule generation^{10,11}. Some studies focused on generating novel and stable 3D molecules^{12–16}. For example, Hoogetboom et al.¹⁴ designed an E(3) equivariant diffusion model, which learned the joint distribution of the continuous atom coordinates and the discrete atom types. Xu et al.¹⁵ proposed a geometric latent diffusion model to map the raw geometric distribution to a lower-dimensional representational space, which can help to overcome the problem caused by the assumption that the discrete variable follows the standard Gaussian distribution. Peng et al.¹⁶

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presented a diffusion model that can generate atoms and bonds simultaneously, while still maintaining their consistency.

Some studies enabled the generation of 3D molecules given the target protein pocket information. For instance, Schneuing et al.¹⁷ developed an equivariant diffusion model for structure-based drug design with two strategies, one is to treat the protein pocket information as a fixed environmental condition, and the other is to model the joint distribution of the protein-ligand complex; Huang et al.¹⁸ trained a dual diffusion model to learn two different virtual edges in molecule graphs, one is localized edges that represent the covalent bonds, and the other is global edges that simulate the van der Waals force. Lee et al.¹⁹ introduced the non-covalent interactions (NCIs) into 3D molecule generation with the improved reliability of generated molecular conformers.

However, current drug design approaches overlook a common modality, which is textual descriptions related to molecules. With the development of natural language processing, more and more studies, especially generative tasks, have benefited from incorporating natural language over the past several years²⁰. Relevant research includes image or video generation given the textual description^{21,22}, the development of chatting bots based on large language models²³, as well as protein sequence or structure design with predefined protein functions²⁴. Textual descriptions can precisely specify a molecule's substructure or properties. Thus, text-guided molecule generation explicitly highlights key characteristics, which is a capability unattainable by text-free generation approaches.

Although there are some approaches working on text-guided molecule generation, these studies are mainly focused on 2D molecular graphs²⁵ or 1D SMILES strings²⁶. As the most natural representation of molecules, 3D molecular graphs include the coordinates, types, and other features of each atom in molecules. The function of a biomolecule is entirely dependent on its 3D conformation²⁷, which makes it important to generate 3D molecules.

While a sequential strategy that generates a 2D molecule from text followed by standard conformer generation might seem like a logical alternative, it suffers from two fundamental limitations. First, it completely severs the 3D molecular assembly from the textual guidance. As conventional 2D-to-3D algorithms like RDKit²⁸ rely exclusively on topological graphs as input, any contextual guidance implied by the original text description are discarded. Second, conventional 3D conformer generators primarily optimize for global minimum free-energy states in a vacuum. However, it is well-established in conformational analyses that the bioactive conformations required for protein binding often deviate significantly from these lowest-energy states due to target-induced reorganization²⁹. Consequently, these approaches frequently fail to capture the specific, functionally relevant 3D geometries.

The textual description of a molecule may contain structural and functional descriptions, thus making it useful to guide 3D molecule generation. However, several challenges are underlined. For instance, the atom coordinates distribution is consistent while the atom types distribution is discrete, which makes it challenging to model these features in parallel. The dynamic nature and diversity of the molecular 3D structures also pose significant challenges to aligning them with the textual descriptions. Another challenge is that the amount of data in the molecular description databases is too small to train a reliable deep learning model, so pretrained language models are necessary to improve the performance³⁰. Although a recent study³¹ attempted to generate 3D molecules from text, they have not considered the connection between textual modalities and molecular 3D structures, which may limit its practical application scenarios and value.

To address the aforementioned challenges, we proposed a Text-guided De novo 3D molecule generation approach based on a multi-modal diffusion model named TDMol. To the best of our knowledge, TDMol is the first method introducing text-3D modality alignment to conditional molecule generation. We first train two contrastive learning models to align the textual descriptions, 2D molecular graphs, and 3D molecular graphs, as well as a prior network to translate textual descriptions to 2D molecular graphs. Then we trained a conditional diffusion model to generate 3D molecules under the guidance of the

textual descriptions. We adopted the equivariant graph neural network in the diffusion model to keep the equivariance of translation, rotation, reflection, and permutation.

The experimental results demonstrate that TDMol generates molecules with valid and reliable 3D structures, outperforming other baseline methods.

Results

Evaluation metrics

In TDMol, there are two main tasks: one is modality alignment, the other is 3D molecule generation. To demonstrate the effectiveness of the modality alignment, we evaluate the CL models in TDMol with the following four metrics:

- a) *Mean Rank*. It is the mean rank of the molecules and calculated as:

$$\text{Mean Rank} = \frac{\sum_{i=1}^n R_i}{n}, \text{ while rank } (R_i) \text{ refers to the position of the true}$$

matching sample in the ranked list of candidates based on cosine similarity in the embedding space. Lower 'Mean Rank' indicates better performance.

- b) *Mean Reciprocal Rank (MRR)*. It is the mean reciprocal rank of the

molecules and calculated as: $MRR = \frac{\sum_{i=1}^n \frac{1}{R_i}}{n}$. Higher 'MRR' indicates better performance.

- c) *Top 1 acc*. It is the percentage that the rank of molecules is lower than 1

and is calculated as: $Top1acc = \frac{\sum_{i=1}^n 1_{R_i \leq 1}}{n}$. Higher "Top 1 acc" indicates better performance.

- d) *Top 10 acc*. It is the percentage that the rank of molecules lower than 10

and calculated as $Top10acc = \frac{\sum_{i=1}^n 1_{R_i \leq 10}}{n}$. A higher "Top 10 acc" indicates better performance.

And we evaluate the generated 3D molecules with the following three metrics:

- a) *Valid (%)*: the percentage of molecules that satisfy the chemical validity measured by RDKit.
 b) *Atom stable (%)*: the percentage of atoms in molecules that maintain chemically valid valency states.
 c) *Mol stable (%)*: the percentage of molecules where all constituent atoms maintain chemically valid valency states.

The "Atom stable" and "Mol stable" metrics reflect the validity of generating molecular 3D structures, with higher values indicating a greater proportion of valid 3D structures within the generated molecules.

We further evaluate the similarity between the generated 3D molecules and both the input text guidance and the corresponding 3D molecules.

- a) *Emb*: the embedding cosine similarity between the generated 3D molecular embedding and the text embedding.
 b) *FPS_Topo*: the topological fingerprint similarity between the generated molecules and the corresponding molecules calculated by RDKit.
 c) *FPS_MACCS*: the MACCS fingerprint similarity between the generated molecules and the corresponding molecules calculated by RDKit.
 d) *FPS_ECFP4*: It is the extended connectivity fingerprint similarity between the generated molecules and the corresponding molecules calculated by RDKit.

These similarity metrics take values within the range [0, 1], with higher numbers indicating greater similarity to the ground truth.

Table 1 | Performance of the CL models and a random control

Method ^a	Mean Rank(_↓)	MRR(_↑)	Top 1 acc(_↑)	Top 10 acc(_↑)
Random ^b	1180	0.004	0.0004	0.0055
Text-2D	3.87	0.83	0.73	0.97
2D-noised 3D	31.70	0.251	0.127	0.519
Text-noised 3D	32.66	0.232	0.112	0.507
Text-noised 3D (direct)	38.21	0.213	0.099	0.472

^a“Text-2D” represents the CL model on textual descriptions and 2D molecular graphs, while “2D-noised 3D” represents the CL model on 2D molecular graphs and noised 3D molecular graphs with timestep T. “Text-noised 3D” denotes the integration of a two-stage CL model, while “Text-noised 3D (direct)” refers to the CL model directly trained to align textual descriptions and noised 3D molecular graphs.

^b“Random” represents that the embeddings for evaluation are randomly generated.

Table 2 | Performance comparison of different methods on the quantitative evaluation metrics over 1000 generated molecules

Method	Valid	Atom stable	Mol stable
EDM	92.6	81.3	/
GeoLDM	99.3	84.4	/
GCDM	/	89.0	5.2
TDmol ^{a,c} (no post-processing ^{b,c})	96.8 ± 0.01	81.7 ± 0.6	/
TDmol	100 ± 0.0	97.3 ± 0.6	46.0 ± 2.7

^aAll metrics of TDmol are averaged over 3 runs.

^bTDmol applied a post-processing strategy to transfer 3D point clouds to molecules. To ensure a fair comparison, we report the metrics for TDmol in two distinct settings: ‘TDmol (no post-processing)’ and ‘TDmol’.

^cUnlike the baseline methods, TDmol is a conditional generative model that utilizes textual descriptions randomly sampled from the test set as input constraints to guide the generation. The results of EDM, GeoLDM, and GCDM are obtained from their papers. TDmol is evaluated under a text-conditioned generation setting, whereas all baseline methods perform unconditional generation. Bold face indicates the method yields the best result across the compared methods.

TDmol enables the modality translation between molecule textual descriptions and the 2D/3D molecular structures

TDmol adopts a two-stage alignment framework: The first stage establishes the correspondence between molecular textual descriptions and 2D features, while the second stage aligns these 2D features with noised 3D molecular conformations perturbed by the diffusion model. The text encoder adopts a SciBERT-based pretrained framework following Text2mol³⁰. While both 2D and 3D molecular encoders share a similar GCN-based architecture, the 3D encoder uniquely incorporates timestep-conditioned embedding layers to handle the timestep-dependent 3D data.

After training the CL model on the benchmark dataset, we evaluated its performance on the test set, with quantitative metrics comprehensively as shown in Table 1. A randomized approach serves as the comparative control, since no established baseline methods currently exist for the task of molecular modality alignment from textual descriptions to 3D representations.

The results demonstrate that our CL models significantly outperform the random control baseline across all evaluation metrics. While the Text-2D model achieves optimal performance metrics, the 2D-noised 3D model exhibits a marginal performance gap. The suboptimal performance of the 2D-noised 3D alignment model could be attributed to the noise-amplification effect in diffusion processes: Higher timesteps correlate with increased noise amplitudes in perturbed conformations, fundamentally

compromising the model’s ability to discriminate geometric signals from stochastic perturbations.

Furthermore, to demonstrate the effectiveness of the two-stage CL model, we trained a direct text-to-3D CL model for comparison. As shown in Table 1, the two-stage approach achieves higher performance metrics. This improvement can be attributed to the introduction of 2D representations as an intermediate bridge, which simplifies the alignment between text and 3D modalities.

TDmol outperforms baseline methods on generating 3D molecules

Building upon the EDM¹⁴ framework, we implemented a text-conditioned molecular generator that integrates cross-modal control through differentiable alignment layers during the denoising process.

While diffusion models typically process 3D molecules as point clouds containing atomic coordinates and types for each atom in the molecules, it should be noted that chemical molecules fundamentally constitute systems composed of atoms connected by chemical bonds with appropriate valence states. Consequently, the conversion from 3D point cloud representations to chemically valid molecular structures may still result in valence state inconsistencies, missing atoms, and other molecular integrity issues. Thus, we have developed a post-processing mechanism that partially corrects errors arising during the conversion of 3D point clouds into chemically valid molecular structures.

We compared our method TDmol with three baseline models EDM¹⁴, GeoLDM¹⁵, and GCDM³² on quantitative evaluation metrics over 1000 generated molecules, as shown in Table 2. While the baseline models perform unconditional generation, TDmol applies additional textual descriptions from the test set to achieve conditional generation. As shown in Table 2, our method demonstrated a 4.2% improvement in ‘Valid’ compared to the EDM baseline. This improvement demonstrates that when using text prompts, the gradient guidance effectively guides the diffusion sampling towards regions of the chemical space that are chemically valid. Furthermore, with the application of post-processing, TDmol achieves 100% ‘Valid’, surpassing all baseline models, including the state-of-the-art GeoLDM (99.3%). However, this improved performance comes at an operational cost that TDmol necessitates a textual input for every generation task. Regarding stability metrics, TDmol outperforms the baselines, including EDM, GeoLDM, and GCDM, in both atom and molecular stability.

TDmol generates molecules with high similarity to the ground truth

To validate the effectiveness of TDmol’s text-conditioned strategy, a comparative analysis is conducted between 3D molecules generated with and without textual description guidance. We utilized all 2,344 textual descriptions from the test set as defined in Section 2.1.1 for conditional generation. For each description, 100 independent samples were generated, resulting in a total of 234,400 molecules. The results are shown in Table 3, where we can observe that the generated molecules with text guidance exhibited improvements across all evaluation metrics compared to those without textual guidance. This improvement demonstrates the efficacy of our conditional generation strategy guided by textual descriptions.

Specifically, the 3D Embedding similarity (Emb) performs a substantial increase from 0.093 to 0.131, indicating that the generated 3D molecular conformations align significantly better with the high-dimensional semantic space of the textual descriptions.

The generated molecules with text guidance demonstrate a significant improvement (40.86%) in the embedding similarity but only a modest enhancement in three fingerprint-based similarity (10.61%, 11.46%, and 28.57%). The potential reason could be that the model directly produces 3D point clouds with features, and errors are introduced when these are converted into molecular structures.

Table 3 | Performance comparison between TDMol generated molecules with text-conditioned and without over 234,400^a samples

Method	Emb (↑)	FPS_Topo (↑)	FPS_MACCS (↑)	FPS_ECFP4 (↑)
No guidance	0.093	0.132	0.419	0.007
Text guidance	0.131	0.146	0.467	0.009

^a All 2,344 textual descriptions in the test are used for conditional generation, with 100 independent samples generated for each description. Bold face indicates the method yields the best result across the compared methods.

Notably, a correlation analysis is performed for the embedding similarity and the fingerprint-based similarity metrics in Table S2. The result confirms a consistent positive correlation between Emb and fingerprint-based metrics, though the strength varies significantly depending on the underlying logic of each fingerprint.

Given the inherent stochasticity in generative models' molecular outputs, introducing a suitable scoring function proves critical for enabling users to discriminate candidate structures. We implement embedding similarity as the model's intrinsic scoring function, where generated molecules having higher cosine similarity with textual description embeddings receive higher scores.

To demonstrate the effectiveness of the text-guided strategy of TDMol, we visualize the distributions of the generated molecules and the ground truth molecules. As shown in Figure S1, 1000 molecules from four categories are visualized by t-distributed stochastic neighbor embedding (t-SNE) simultaneously³³. We further quantitatively measured the distances between these four distributions by fitting the t-SNE results to bivariate normal distributions, as shown in Fig. 1.

In Fig. 1A, the general formula of the bivariate normal distribution is given in the first row, and the mean μ and covariance Σ of the four distributions are listed below. The text guidance distribution has the closest mean and covariance to the ground truth distribution. These four fitted bivariate normal distributions are visualized with the probability density levels in Fig. 1B. In Fig. 1C, the distance of two distributions is defined as the Euclidean distance between their fitted means. The text guidance distribution achieves the minimum distance of 0.222, which indicates that it is closest to the ground truth distribution.

Additionally, to evaluate the chemical properties of the generated structures, we plotted the distributions of QED and SAscore in Figure S2.

Figure 2 shows a set of examples to visually illustrate the 3D embedding similarity and the molecular fingerprint similarity among the 3D molecules generated by TDMol with no guidance, the molecules generated by TDMol with text guidance, and the ground truth molecules corresponding to the given textual descriptions. TDMol demonstrates the capability to generate molecules with high similarity to the ground truth.

As shown in Fig. 2, molecules generated with text guidance exhibit higher 3D embedding similarity than those without guidance. Moreover, we can observe that the 3D embedding similarity and the fingerprint similarity do not always align. In some cases, molecules with high 3D embedding similarity exhibit low fingerprint similarity. This discrepancy arises because molecular fingerprints typically focus more on capturing 2D substructural features, whereas 3D embeddings learn representations from the overall 3D molecular structures.

User-friendly online webserver of TDMol

To enhance the accessibility of our proposed method for the broader research community, we have developed a user-friendly online webserver of TDMol for academic use at <http://www.csbio.sjtu.edu.cn/bioinf/TDMol>. This platform allows researchers to generate 3D molecular structures directly from textual descriptions without requiring local computational resource.

The webserver offers a streamlined and intuitive workflow for molecular generation, as shown in Figure S5. To initiate the process, users can provide a natural language description of the desired molecules (subject to a

minimum length of 20 characters to ensure sufficient semantic context). Upon submission, the backend conditional diffusion model processes the text prompt and generates a valid 3D molecular conformation in approximately 30 seconds.

The generated molecules are visualized with the corresponding SMILES strings and 2D structures in the result interface. We also integrated the JSmol³⁴ visualization tool, enabling users to interactively inspect the generated 3D geometries directly within the browser. Furthermore, the online webserver includes an automated similarity search module that retrieves and links to the most structurally similar compounds to the generated molecules in the ChEMBL database. For each generated molecule, this module calculates a topological fingerprint with a dimension of 2048 by RDKit and performs a similarity search against the ChEMBL database. The top 10 most similar bioactive molecules are displayed in the detailed result pages of the generated molecules. Detailed information and external links to the ChEMBL entries are provided, allowing users to quickly compare the generated candidates with known drugs or bioactive compounds.

This webserver effectively bridges the gap between advanced generative AI models and practical drug discovery workflows, enabling real-time conceptualization and validation of novel molecules.

Methods

Benchmark datasets

We adopt the ChEBI-20³⁰ dataset for text-2D molecule alignment. There are 33,010 text-molecule pairs collected from the ChEBI database, filtered based on textual descriptions exceeding 20 words. For the 2D-3D molecule alignment, we first collect the 3D conformers of these molecules from PubChem database. Conformer generation in PubChem for some molecules is disallowed due to too many atoms; thus, this dataset is a subset of ChEBI-20 containing 23,182 molecules with their 3D structures.

Then, these 3D molecular conformations subsequently undergo the forward diffusion process (a progressive noise injection), generating noised 3D structural representations to build a benchmark dataset for 2D-noised 3D molecule alignment. These noised 3D structural representations are timestep-dependent, where the timestep T functions as a critical hyperparameter in the diffusion model, typically configured with $T = 1000$ iterations. Since excessively large T values drive the noised data towards standard Gaussian noise, we empirically set the timestep range to $T \in [0, 200]$ for perturbed data generation. This implementation yields 200 corresponding 2D-noised 3D pairs per original 2D-3D pair. We then randomly split these data into the training, validation, and test sets with a ratio of 8:1:1. The textual descriptions and the molecular properties distribution in the training dataset are visualized in Fig. 3.

We select GEOM-Drug³⁵ dataset to train the diffusion model for 3D molecule generation. GEOM-Drug contains high-quality conformers consisting of 317,928 mid-sized organic molecules with experimental data. Following EDM's setting¹⁴, we retained 30 conformers with the lowest energy for each molecule. We randomly split these data into the training, validation, and test sets with a ratio of 8:1:1.

Two-state modality alignment with contrastive learning

Contrastive learning (CL) models can extract meaningful embeddings of the data, assuming that the similarity distance of two samples should be close in the embedding space if they share the same label, and be apart if they have

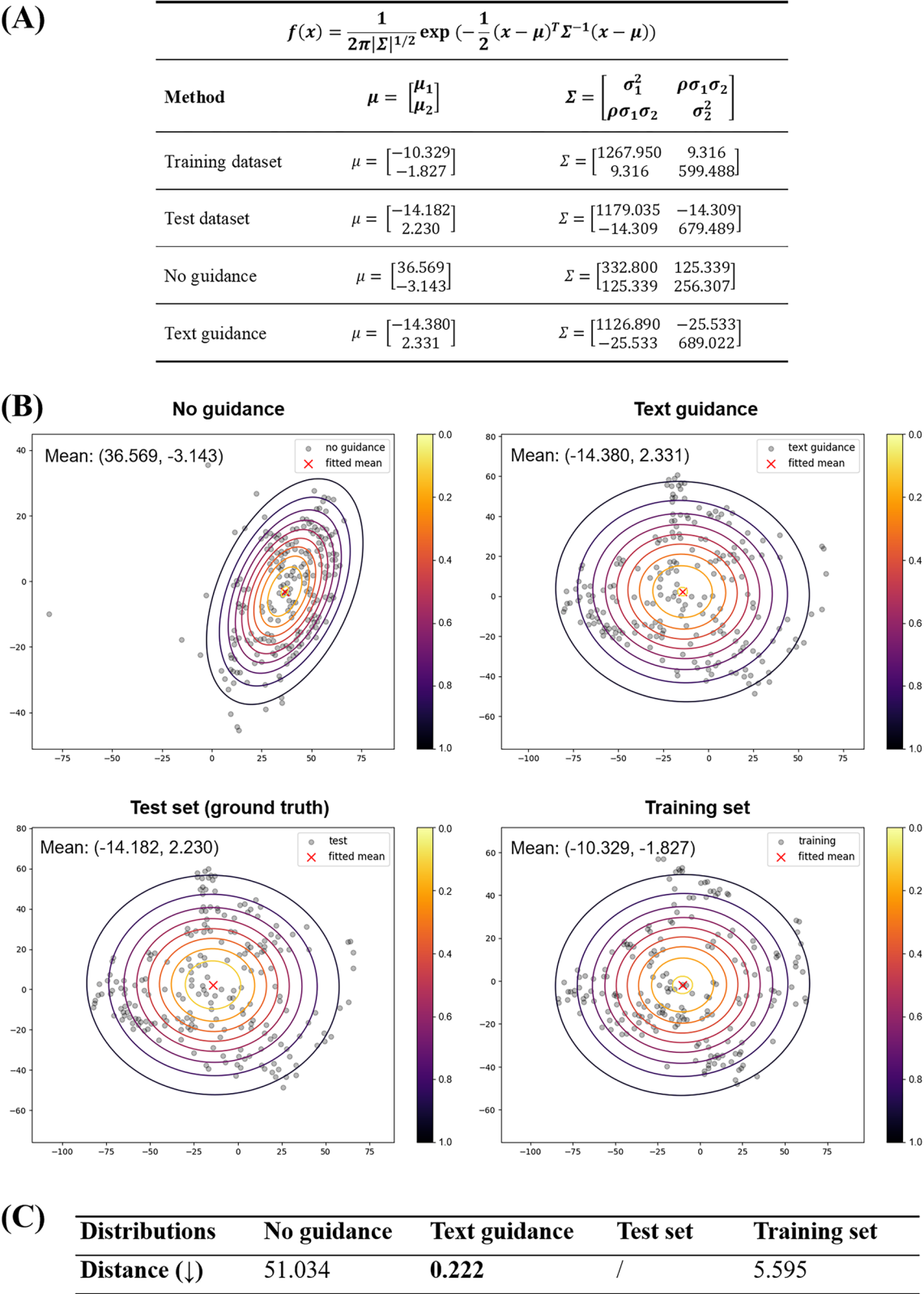


Fig. 1 | The fitted bivariate normal distributions from t-SNE results. **A** The detailed formulas of the four distributions. **B** The visualization of the four distributions. The contour lines of different colors in the figure represent distinct probability density levels. **c** The distances between the distributions and the ground truth distribution.

different labels. CL models can also be used to align different modalities of the same sample. Since it is challenging to directly align noisy 3D molecular data with the textual descriptions, we introduced the 2D molecular graph as an intermediate bridge, decomposing the 3D-text alignment model into two CL models. One is to align the 2D structure of the molecules and their textual descriptions, and the other is to align the 2D structure of the molecules and the corresponding 3D structures noisy by the diffusion forward process, as shown in Fig. 4. The 2D molecular graphs are noise-free

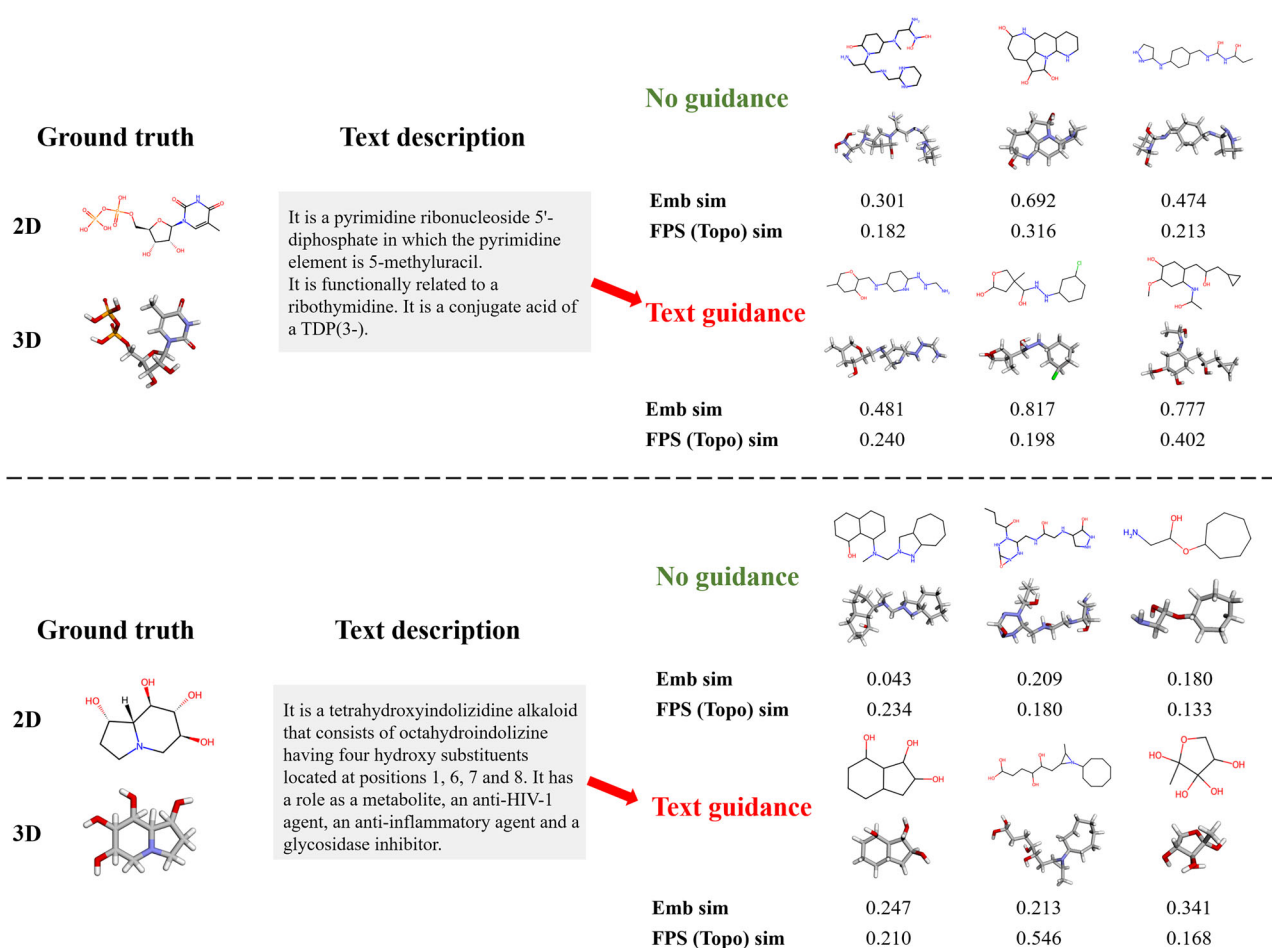


Fig. 2 | Molecules generated by TDmol with and without text guidance. The input textual descriptions and the corresponding ground truth molecules are listed in the left. The 3D embedding similarity and the topological fingerprint similarity scores of the generated molecules are provided at the bottom.

which enables effective alignment with the textual descriptions, while the shared features containing “number of atoms” and “atom types” between 2D and 3D molecular graphs act as structural anchors that significantly reduce the complexity of the cross-modal alignment.

The pretrained sci-BERT model is chosen to be the model structure of the text encoder; the molecular 2D encoder and the molecular 3D encoder are built with almost the same GNN blocks, while the molecular 3D encoder has additional timestep embedding layers to receive the timestep T as input, the detailed model architectures are shown in Figure S3. The training loss is set to be the addition of two cross-entropy losses which follows CLIP’s setting³⁶, as shown below.

$$CL\ Loss1 = CE(SIM(x, h), label)_{axis=0} + CE(SIM(h, x), label)_{axis=1} \quad (1)$$

$$CL\ Loss2 = CE(SIM(h, k), label)_{axis=0} + CE(SIM(k, h), label)_{axis=1} \quad (2)$$

where $SIM(x, h)$ calculates the cosine similarities of the text embedding x and the molecular 2D embedding h . The “axis” refers to the dimension of the similarity matrix. In the formula 1, $axis = 0$ denotes text-to-molecule alignment (classifying the correct molecule given a text input), while $axis = 1$ denotes molecule-to-text alignment (classifying the correct text given a molecule). Since the data consists of paired samples, the $label$ for the i -th sample in the first modality is simply the index i in the second modality. This means the ground truth targets are the diagonal elements of the similarity matrix.

Denoising diffusion probabilistic model

Diffusion models are deep generative models that learn the data distribution through a forward diffusion process and a reverse denoising process.

The forward diffusion process is a Markov process, in which random Gaussian noises z are gradually added to the data $x_0 \sim q(x)$ from timestep $t = 0$ to T , as shown in the following formula.

$$q(x_t | x_{t-1}) = N(x_t; \alpha_t x_{t-1}, \sigma_t^2 I) \quad (3)$$

$$q(x_{0,1,2,\dots,T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}) \quad (4)$$

where α_t and σ_t^2 represent the weight of signal and noise, respectively.

Denoising diffusion probabilistic model (DDPM)³⁷ designed a signal-noise schedule as:

$$\alpha_t^2 + \sigma_t^2 = 1 \quad (5)$$

As t approaches 1, α_t approaches 1, σ_t^2 approaches 0, making the distribution of $q(x_t)$ close to the raw data distribution; and when t approaches infinity, α_t approaches 0, σ_t^2 approaches 1, making the distribution of $q(x_t)$ close to Gaussian noises.

The reverse denoising process aims to reconstruct the data distribution $p_\theta(x_0)$ from the gaussian noises $p(x_T)$ by deep neural networks with

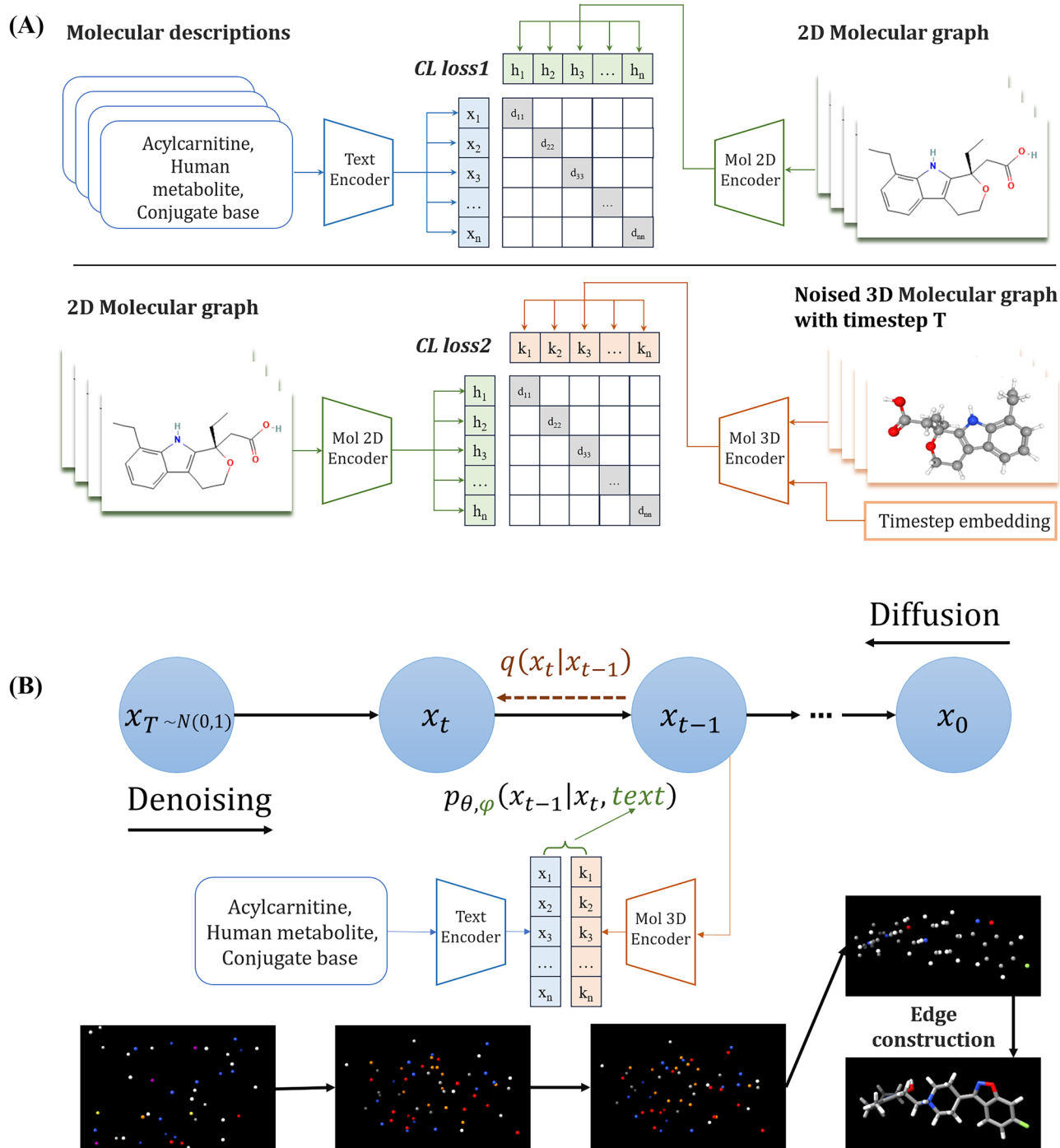


Fig. 4 | The workflow of the proposed TDmol approach for text-guided 3D molecule generation. **A** Proposed two-stage contrastive learning model structure. The text encoder adopts a model architecture similar to SciBERT⁴², while both 2D and 3D molecular encoders share a similar GCN-based architecture, and the 3D encoder uniquely incorporates timestep-conditioned embedding layers. First, the text encoder and the 2D molecular encoder are trained to align molecular

descriptions and 2D molecular graphs. Then the 3D molecular encoder is trained under the limitation that the layers of 2D molecular encoder are frozen. **B** Text-guided sampling based on diffusion. The gradients from the contrastive learning model are leveraged as guidance for molecule generation, which manifests as modifying the sampling distribution of the diffusion model by the shifted mean.

Classifier-guided conditional diffusion model

In the section 2.3, we explained the basic algorithm of the unconditional diffusion model, and when there is an additional guidance y and a trained classifier $p_{\phi}(y|x_t, t)$, the conditional distribution in formula 7 can be written as:

$$p_{\theta, \phi}(x_{t-1}|x_t, y) = Z \cdot p_{\theta}(x_{t-1}|x_t) p_{\phi}(y|x_{t-1}, x_t) \quad (13)$$

where Z is a normalization constant.

In the diffusion process, x_t is derived from x_{t-1} by adding independent Gaussian noise in the forward direction. Therefore, x_{t-1} contains the structural signal present in x_t but contains less noise. We assume that conditioned on x_{t-1} , the additional noise component in x_t provides no further information about the semantic label y . In other words, x_{t-1} is a

sufficient statistic for y relative to x_t . So the formula 14 can be rewritten as:

$$p_{\theta,\varphi}(x_{t-1}|x_t, y) = Z \cdot p_{\theta}(x_{t-1}|x_t) p_{\varphi}(y|x_{t-1}) \quad (14)$$

It is proven that the conditional distribution can be approximated by a Gaussian distribution with a shifted mean^{39,40}.

$$\begin{aligned} \log(p_{\theta}(x_{t-1}|x_t) p_{\varphi}(y|x_{t-1})) &\approx \log p(z) + C, \\ z &\sim N(z; \mu_{\theta} + \sigma_{\theta}^2 I g, \sigma_{\theta}^2 I) \end{aligned} \quad (15)$$

where $g = \nabla_{x_{t-1}} \log p_{\varphi}(y|x_{t-1})$. y can be a discrete class label, $p_{\varphi}(y|x_{t-1})$ represents the probability that x_{t-1} belongs to class y , and g can be calculated as the gradient of the classifier given x_{t-1} .

In this study, we trained a CL model as described in section 2.2 to substitute the classifier, and the text modality of the molecules to substitute the discrete label y . The g can be written as:

$$g = \nabla_{x_{t-1}} \text{Encoder}_{mol}(x_{t-1}, t-1) \cdot \text{Encoder}_{text}(text) \quad (16)$$

where Encoder_{mol} and Encoder_{text} represent the two encoders of the trained contrastive learning model. The conditional generation process is described in Algorithm 1, and the sampling pipeline is shown in Fig. 4B.

Algorithm 1. Text-guided molecule generation by TDmol

Step 1. Train unconditional diffusion model and perturbed data generation

Input: Data x from GEOM-Drug, neural network with parameters θ , Data x_{raw} from ChEBI-20

Sample $t \sim [0, T]$, $\epsilon \sim N(0, I)$

Minimize $\|\epsilon - \epsilon_{\theta}(\sqrt{\alpha_t}x_0 + \sqrt{1-\alpha_t}\epsilon, t)\|^2$

For $t = 1 \dots 200$ **Do**

calculate $x_t = \sqrt{\alpha_t}x_{raw} + \sqrt{1-\alpha_t}\epsilon$

End For

Return (x_t, t) // sampling noised data from the diffusion model

Step 2. Train the two-stage contrastive learning models.

Input: text description x_{text} , 2D molecular graph x_{2D} , two neural networks Encoder_{text} , Encoder_{2D}

Freeze Encoder_{text}

Calculate $j = \text{Encoder}_{text}(x_{text})$, $h = \text{Encoder}_{2D}(x_{2D})$

Minimize $CE(\text{SIM}(j, h)) + CE(\text{SIM}(h, j))$ // symmetric cross-entropy loss

Return Encoder_{2D} // text-2D alignment

Input: paired 3D molecular data (x_t, t) , 2D molecular graph x_{2D} , two neural networks Encoder_{3D} , Encoder_{2D}

Freeze Encoder_{2D}

Calculate $h = \text{Encoder}_{2D}(x_{2D})$, $k = \text{Encoder}_{3D}(x_t, t)$

Minimize $CE(\text{SIM}(h, k)) + CE(\text{SIM}(k, h))$

Return Encoder_{3D} // 2D-3D alignment

Step 3. Sample molecules with text guidance.

Input: text guidance $text$, scaling factor s // s is used to control the degree of text influence during generation

Given: diffusion model $(\mu_{\theta}, \sigma_{\theta})$, CL model $\text{Encoder}_{3D}(x_t, t)$, $\text{Encoder}_{text}(text)$ sample $x_T \sim N(0, I)$

For $t = T \dots 1$ **Do**

diffusion sampling (μ, Σ) from $(\mu_{\theta}, \sigma_{\theta}^2 I)$

calculate gradient $g = \nabla_{x_t} \text{Cos}(\text{Encoder}_{3D}(x_t, t), \text{Encoder}_{text}(text))$, where

$$\text{Cos}(a, b) = \frac{a \cdot b}{\|a\| \|b\|}$$

sample x_{t-1} from $N(\mu + sg, \Sigma)$

End For

Return x_0

Equivariant graph neural networks

Given a transformation set G and an element $\tau \in G$, a function φ is defined to be equivariant to τ when $\varphi(\tau(x)) = \tau(\varphi(x))$. Neural networks can be designed to be learnable equivariant functions for transformations including translation, rotation, etc. To keep the equivariance, we adopted the equivariant graph neural networks (EGNNs)⁴¹. In this study, chemical molecules are encoded as molecular graphs by treating atoms as nodes. Specifically, we build fully connected molecular graphs G_m where every two atoms $v_i, v_j \in V$ form an edge. The node features consist of the atom coordinates $x \in \mathbb{R}^3$ and the one-hot encoding of the atom types h . Equivariant graph convolutional layers (EGCLs) are the basic units of EGNN, which take the node features of the l -th layer h^l and x^l as input and output the node features of the next layer h^{l+1} and x^{l+1} as follows:

$$m_{ij} = \phi_e(h_i^l, h_j^l, d_{ij}^2, a_{ij}) \quad (17)$$

$$m_{ij} = \phi_e(h_i^l, h_j^l, d_{ij}^2, a_{ij}) \quad (18)$$

$$x_i^{l+1} = x_i^l + \sum_{j \neq i} \frac{x_j^l - x_i^l}{d_{ij} + 1} \phi_x(h_i^l, h_j^l, d_{ij}^2, a_{ij}) \quad (19)$$

where d_{ij} represents the Euclidean distance between the atom i and j , a_{ij} represents the additional edge information, ϕ_e , ϕ_h , ϕ_{att} , and ϕ_x represent the learnable MLPs.

Discussion

In this study, we introduce TDmol, a multimodal generative approach that bridges molecular textual descriptions with 3D structural generation. TDmol demonstrates the feasibility of leveraging text-based guidance for accurate 3D molecule design. The promising performance of TDmol stems from several design choices: (1) TDmol applies a CL method to align different molecular modalities. The CL models perform accurate alignment between molecular textual descriptions and the 2D/3D molecular graphs with the assistance of the pretrained language model. (2) TDmol adopts a classifier-guided conditional diffusion model to generate 3D molecules. The text guidance is converted to a shifted mean guided by the gradients of the CL models. (3) TDmol proposes a post-processing edge construction method that can ensure that the generated molecules are 100% valid and nearly 100% atom stable.

Despite these advances, several challenges remain, many of which reflect broader issues in multimodal AI for digital medicine. A primary limitation is data scarcity—the current lack of large-scale, high-quality molecular text-structure paired datasets may restrict the generalization capacity of cross-modal alignment models. For instance, the ChEBI-20 dataset, though the largest available corpus of molecular descriptions, contains only tens of thousands of entries. If with larger dataset, the model robustness and scalability are expected to be further improved. Furthermore, errors are introduced during the transitions from the text modality to 2D and from 2D to 3D.

TDmol's current inference speed of ~30 seconds per molecule (unbatched) could be a limitation for applications, primarily due to the complex iterative denoising required for 3D spatial optimization. Future work will address this by integrating accelerated diffusion sampling techniques, such as flow matching models, and implementing multi-GPU parallelized computing pipelines to significantly enhance overall generation throughput.

While end-to-end text-to-3D training is conceptually appealing, its performance remains hampered by the insufficient multimodal training data. Attempting to learn both 3D geometric distributions and semantic alignment simultaneously on limited paired samples often leads to overfitting or mode collapse. Consequently, the two-stage strategy adopted in this work offers a solution by decoupling these tasks: it allows the diffusion backbone to learn physical priors from large-scale unpaired datasets like GEOM-Drug, thereby reserving the scarce text-molecule pairs specifically

for the cross-modal alignment. Interestingly, we observed that moderate noise in early sampling stages may enhance molecular diversity, whereas over-optimized alignment tends to reproduce existing structures, highlighting a trade-off between diversity and similarity.

Another critical issue is the lack of standardized multimodal evaluation metrics. As shown in Table 3, we observe that although the embedding similarity between the generated molecules and the ground truth improves significantly, the molecular fingerprint similarity does not show a notable increase. A potential reason is that molecular fingerprints are descriptors encoded based on 2D molecular substructural features. Since TDMol generates continuous 3D atomic coordinates rather than discrete molecular graphs, calculating 2D fingerprints requires reconstructing bond connectivity and bond orders from these coordinates. This reconstruction step is inherently lossy, suggesting that more powerful metrics are necessary to evaluate the similarity between the generated 3D molecules and the textual descriptions.

Moreover, commonly used metrics for molecule generation, such as uniqueness and novelty, are primarily defined based on 2D structural information, which do not fully capture the novelty and diversity of 3D conformations. A proper evaluation metric on the 3D structural information will be helpful in the future development of 3D conformation generation.

Looking forward, TDMol opens several avenues for future research in multimodal digital medicine. Based on this research, there remain many promising directions for future exploration. For instance, incorporating protein binding pocket information as an additional structural condition brings interaction information, enabling target-aware molecule generation. Moreover, integrating dynamic structural information such as protein flexibility could further improve the functional relevance.

Data availability

The datasets used in this study are available at: <https://github.com/yifang000/TDMol>.

Code availability

Software and Code are available at: <https://github.com/yifang000/TDMol>, www.csbio.sjtu.edu.cn/bioinf/TDMol.

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Author contributions

Y.F.: Conceptualization, Methodology, Writing - Original Draft; Y.L. & Y.X.C.: Investigation, Formal Analysis, Visualization; Z.Y.D.: Software, Visualization; X.Y.P. & H.B.S.: Resources, Supervision, Writing - Review and Editing.

Competing interests

The authors declare no competing interests.

Additional information

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